

CompSci 234/NetSys 210
Advanced Topics in Networking
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Unit 3: Cloud and Data Centers

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Slide adopted from Prof. Gunes' materials

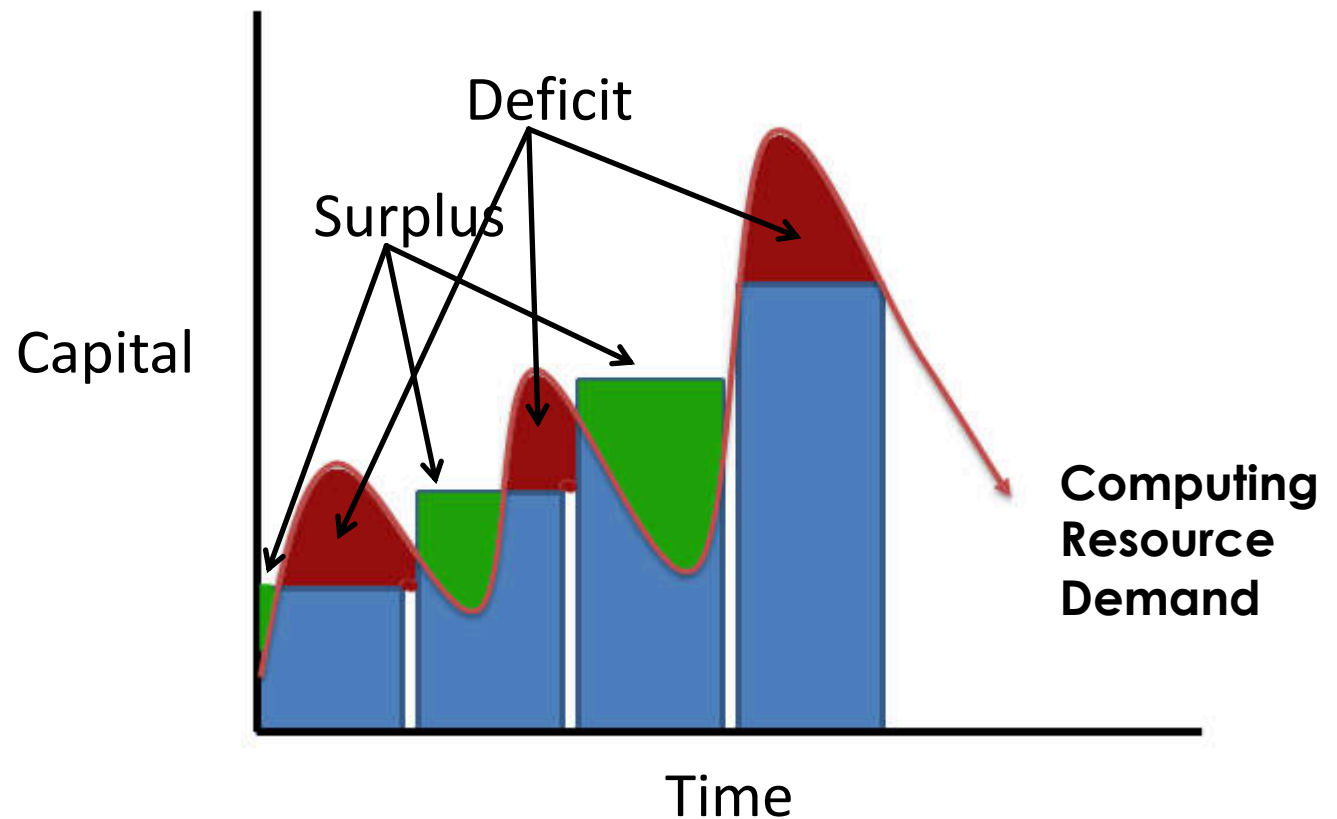
Agenda

- What is Cloud
- Cloud Service Models
- Deployments
- Advantages
- Challenges and Potential Solutions

What is a Data Center

- a centralized location where computers, storages, and networks are collocated
- users need not be physically at the data center to use its resources
- Enterprise data centers appear before cloud computing
 - mainframe (60')
 - minicomputer (80')
 - Amazon's enterprise datacenter (early 2000's), which eventually become a pioneering **public cloud service** (AWS) in 2005

Problem With (Small) Enterprise Data Center



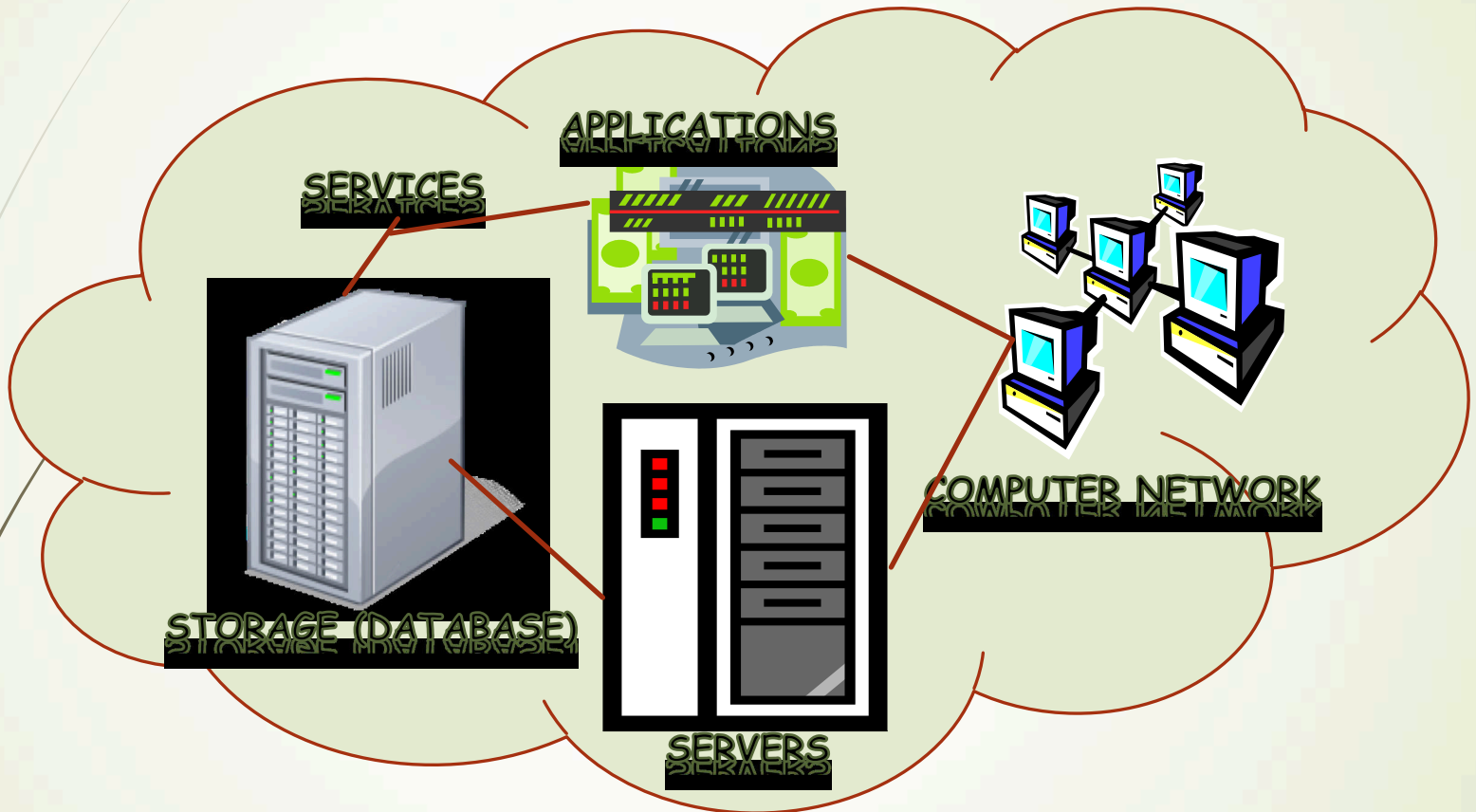
What is Cloud Platform?

- a new class of network based computing that takes place over the Internet
 - was referred to as utility computing earlier
 - **cloud platform**: a bunch of integrated and networked hardware, software and Internet infrastructure
 - delivers hardware, software and networking services over the Internet to clients
- Cloud platforms hide the complexity and details from users and applications by providing simple API or even graphical interface

Cloud Services

- cloud platform provides **on-demand services**
 - always on
 - anywhere, anytime, and anyplace.
- **elastic**: pay for use and as needed,
 - scale up/down in capacity and functionalities
- **hardware** and **software** services are shared among
 - corporations, enterprises, business markets, and general public

Cloud Infrastructure



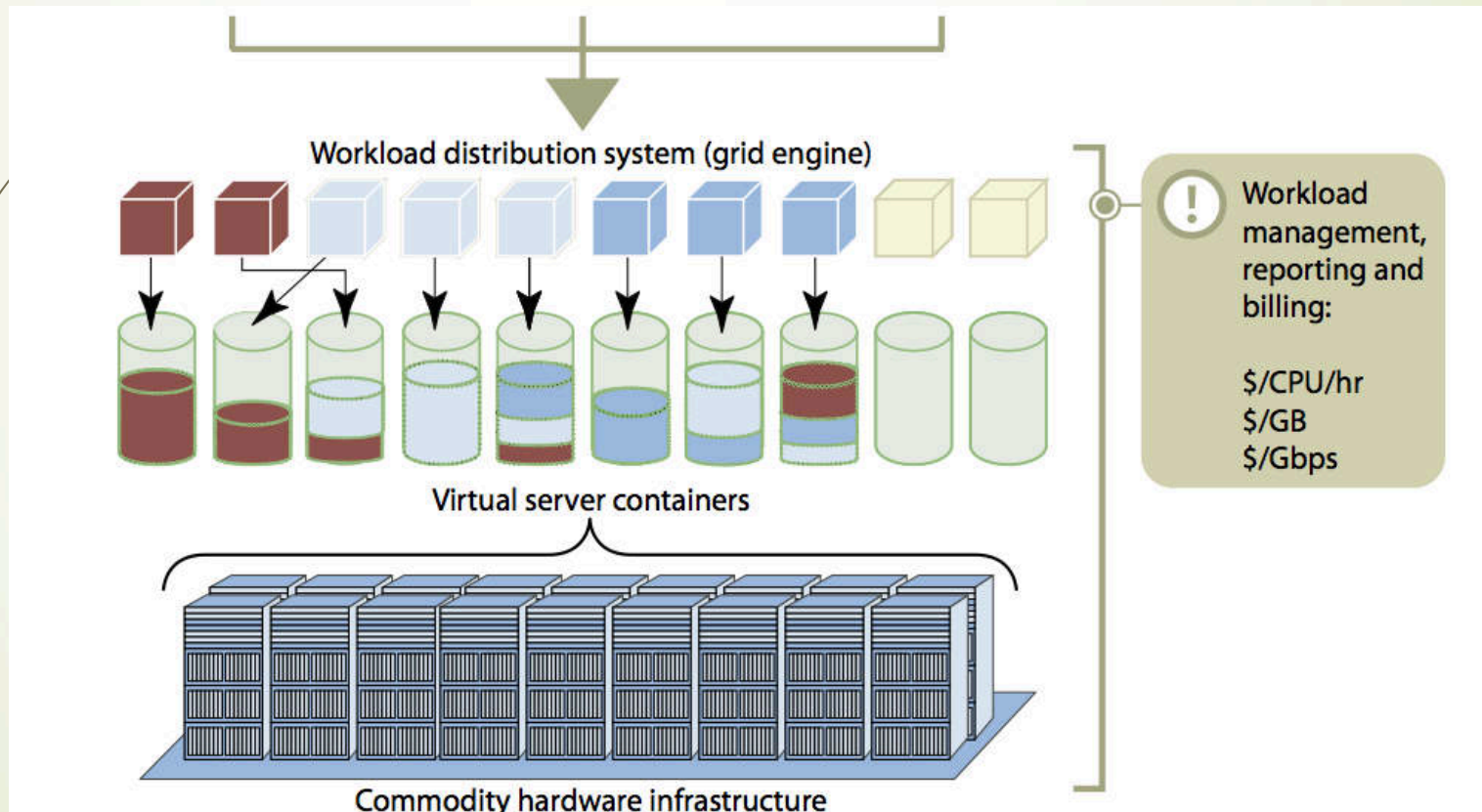
- Shared pool of configurable computing resources
- On-demand network access
- Provisioned by the Service Provider

Key Characteristics of Cloud Computing

- **Remotely** hosted: Services or data are hosted on remote infrastructure
- **Ubiquitous**: Services or data are available from anywhere
- **Commodified**: The result is a utility computing model similar to traditional that of traditional utilities, like gas and electricity - you pay for what you would want!

Serving Multiple Tenants Using Commodity Hardware

Different Cloud Applications (From Different Tenants)



Detailed Characteristics

Common Characteristics

Massive Scale

Resilient Computing

Homogeneity

Geographic Distribution

Virtualization

Service Orientation

Low Cost Software

Advanced Security

Essential Characteristics

On Demand Self-Service

Broad Network Access

Rapid Elasticity

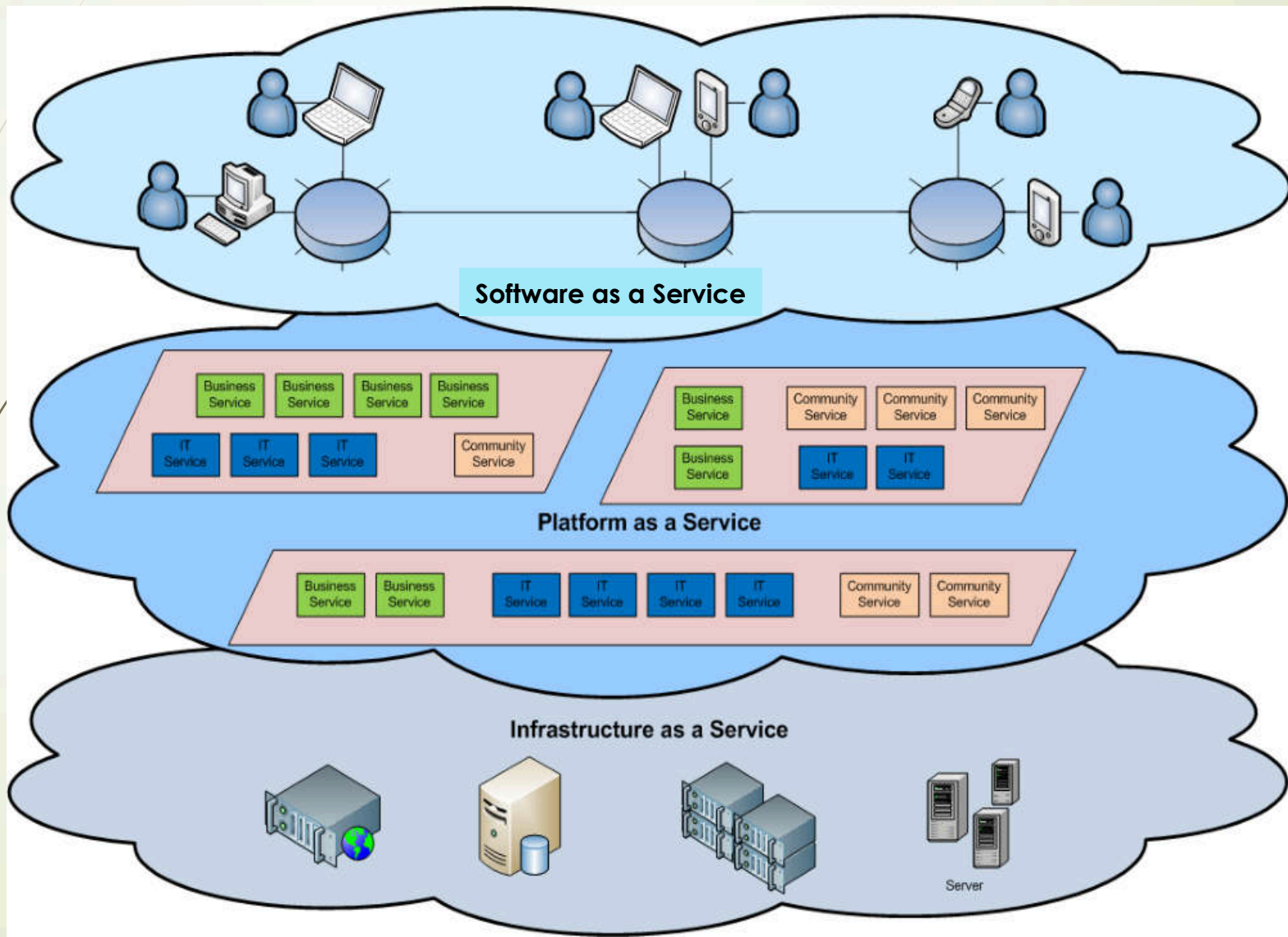
Resource Pooling

Measured Service

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Cloud Architecture and Three Service Models



Service Models

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Software as a Service (SaaS)

Platform as a Service (PaaS)

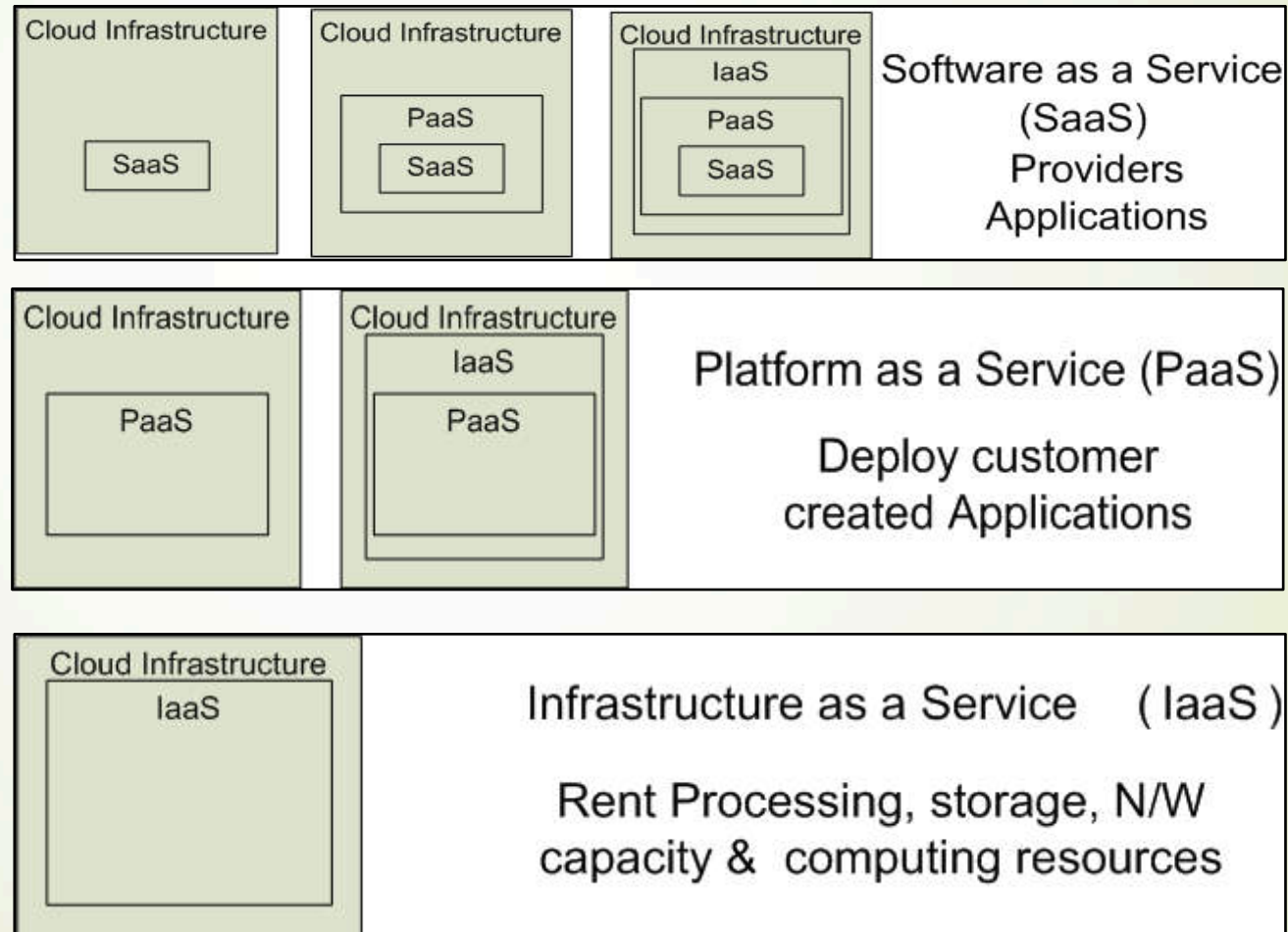
Infrastructure as a Service (IaaS)

SalesForce CRM

LotusLive



Google App



Infrastructure as a Service

- IaaS delivers technology infrastructure as an on demand, scalable, service
 - Usually billed based on usage
 - Usually multi-tenant virtualized environment
 - Can be coupled with Managed Services for OS and application supportw

Platform as a Service

- ▶ PaaS supports the complete life cycle of **building and delivering web applications and services** entirely from the Internet
 - ▶ Typically applications are developed with a particular platform in mind
 - ▶ Multi-tenant environments
 - ▶ Highly scalable multi-tier architecture

Software as a Service

- SaaS provides multi-tenant access to software as a remote Web-based service
 - Usually billed based on usage
 - Usually multi-tenant environment
 - Highly scalable architecture
- SaaS alleviates the burden of software maintenance/support
 - but users relinquish control over software versions and requirements

Cloud Revenue Distribution

